INCH-POUND

MIL-DTL-55302/91E
18 February 2005
SUPERSEDING
MIL-C-55302/91D(CR)
12 February 1982

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, STRAIGHT-THRU, HERMAPHRODITIC CONTACTS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

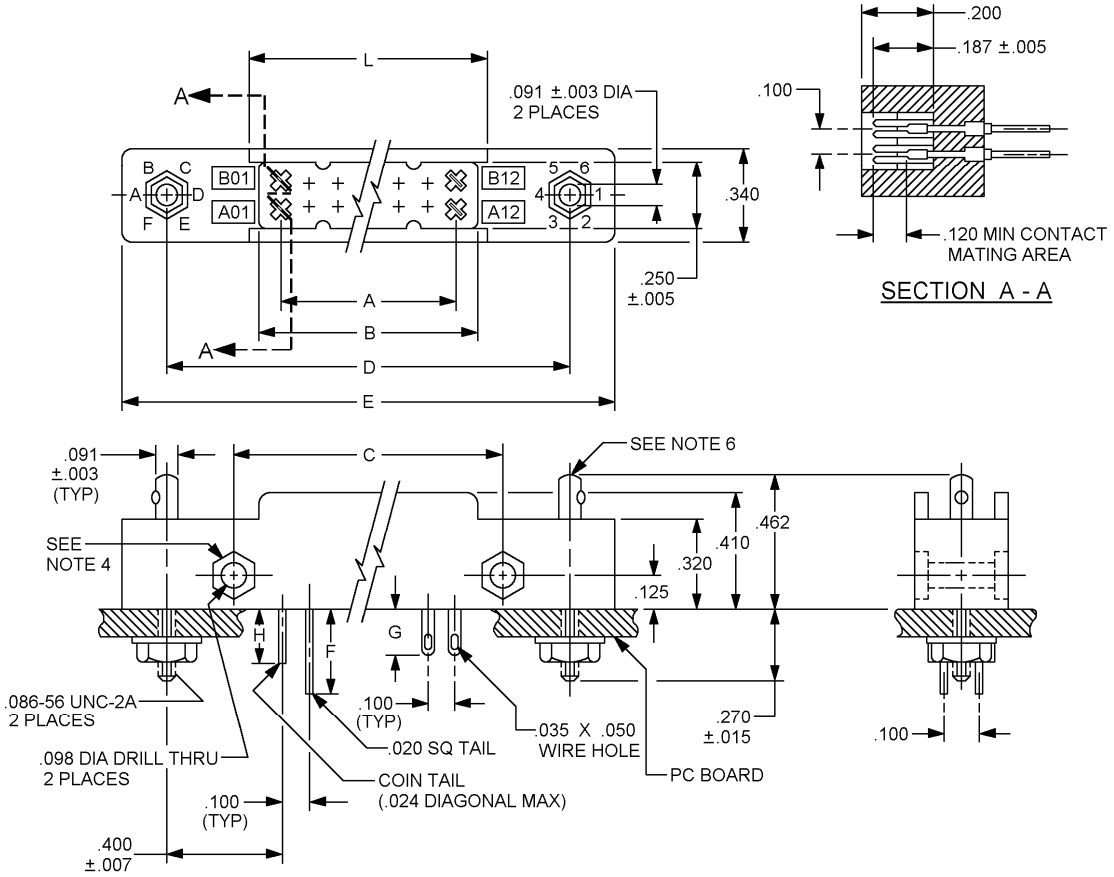
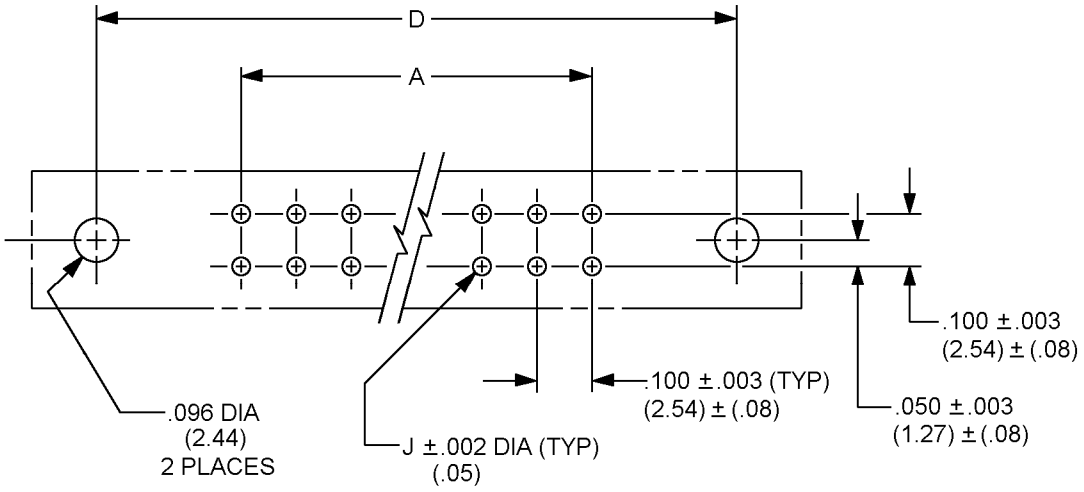
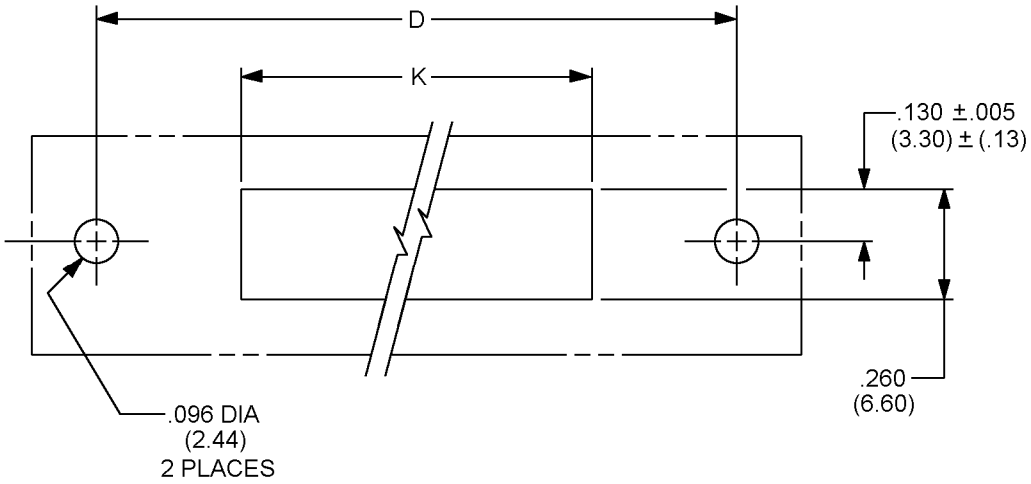


FIGURE 1. Connector, plug (.100 inch spacing).

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TYPICAL HOLE GRID LAYOUT USING P.C. TAIL CONTACT



TYPICAL HOLE CUTOUT USING WIRE HOLE CONTACT

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.032	0.81	.100	2.54	.270	6.86
.005	0.13	.070	1.78	.125	3.18	.320	8.13
.015	0.38	.086	2.18	.187	4.75	.340	8.64
.020	0.51	.091	2.31	.200	5.08	.410	10.41
.024	0.61	.098	2.49	.250	6.35	.462	11.83

FIGURE 1. Connector, plug (.100 inch spacing) – Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .010 (0.25 mm).
4. Dimensions are .162 (4.11 mm) hex across the flats by .052 deep (1.32 mm) – 4 places.
5. Metric equivalents are in parentheses or in columns.
6. Keying sockets shown are in positions 1 and A.

FIGURE 1. Connector receptacle (.100 inch spacing) – Continued.

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TABLE I. Dimensions. 1/

Dash number	Contacts	Dimensions										
		A ±.006 (0.15)	B ±.006 (0.15)	C ±.006 (0.15)	D ±.006 (0.15)	E	F ±.015 (0.38)	G ±.015 (0.38)	H ±.015 (0.38)	J	K	L ±.006 (0.15)
01 2/	24	1.100 (27.94)	1.250 (31.75)	1.400 (35.56)	1.900 (48.26)	2.200 (55.88)	-	-	.400 (10.16)	.026 (0.66)	1.260 (32.00)	1.252 (31.80)
02 3/							-	.162 (4.11)	-	-		
03 4/							.103 (2.62)	-	-	.032 (0.81)		
04 4/							.228 (5.79)	-	-	.032 (0.81)		
05 4/							.259 (6.58)	-	-	.032 (0.81)		
06 4/							.462 (11.73)	-	-	.032 (0.81)		
07 4/							.541 (13.74)	-	-	.032 (0.81)		
29 2/							-	-	.279 (7.09)	.026 (0.66)		
30 2/							-	-	.309 (7.85)	.026 (0.66)		
31 2/							-	-	.341 (8.66)	.026 (0.66)		
32 2/							-	-	.479 (12.17)	.026 (0.66)		
33 2/							-	-	.509 (12.93)	.026 (0.66)		
34 2/							-	-	.541 (13.74)	.026 (0.66)		
08 2/	48	2.300 (58.42)	2.450 (62.23)	2.600 (66.04)	3.100 (78.74)	3.400 (86.36)	-	-	.400 (10.16)	.026 (0.66)	2.460 (62.48)	2.452 (62.28)
09 3/							-	.162 (4.11)	-	-		
10 4/							.103 (2.62)	-	-	.032 (0.81)		
11 4/							.228 (5.79)	-	-	.032 (0.81)		
12 4/							.259 (6.58)	-	-	.032 (0.81)		
13 4/							.462 (11.73)	-	-	.032 (0.81)		
14 4/							.541 (13.74)	-	-	.032 (0.81)		
35 2/							-	-	.279 (7.09)	.026 (0.66)		
36 2/							-	-	.309 (7.85)	.026 (0.66)		
37 2/							-	-	.341 (8.66)	.026 (0.66)		
38 2/							-	-	.479 (12.17)	.026 (0.66)		
39 2/							-	-	.509 (12.93)	.026 (0.66)		
40 2/							-	-	.541 (13.74)	.026 (0.66)		

See footnotes at the end of the table.

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TABLE I. Dimensions – Continued. 1/

Dash number	Contacts	Dimensions										
		A ±.006 (0.15)	B ±.006 (0.15)	C ±.006 (0.15)	D ±.006 (0.15)	E	F ±.015 (0.38)	G ±.015 (0.38)	H ±.015 (0.38)	J	K	L ±.006 (0.15)
15 2/	72	3.500 (88.90)	3.650 (92.71)	3.800 (96.52)	4.300 (109.22)	4.600 (116.84)	-	-	.400 (10.16)	.026 (0.66)	3.660 (92.96)	3.652 (92.76)
16 3/							-	.162 (4.11)	-	-		
17 4/							.103 (2.62)	-	-	.032 (0.81)		
18 4/							.228 (5.79)	-	-	.032 (0.81)		
19 4/							.259 (6.58)	-	-	.032 (0.81)		
20 4/							.462 (11.73)	-	-	.032 (0.81)		
21 4/							.541 (13.74)	-	-	.032 (0.81)		
41 2/							-	-	.279 (7.09)	.026 (0.66)		
42 2/							-	-	.309 (7.85)	.026 (0.66)		
43 2/							-	-	.341 (8.66)	.026 (0.66)		
44 2/							-	-	.479 (12.17)	.026 (0.66)		
45 2/							-	-	.509 (12.93)	.026 (0.66)		
46 2/							-	-	.541 (13.74)	.026 (0.66)		
22 2/	96	4.700 (119.38)	4.850 (123.19)	5.000 (127.00))	5.500 (139.70)	5.800 (147.32)	-	-	.400 (10.16)	.026 (0.66)	4.860 (123.44)	4.852 (123.24)
23 3/							-	.162 (4.11)	-	-		
24 4/							.103 (2.62)	-	-	.032 (0.81)		
25 4/							.228 (5.79)	-	-	.032 (0.81)		
26 4/							.259 (6.58)	-	-	.032 (0.81)		
27 4/							.462 (11.73)	-	-	.032 (0.81)		
28 4/							.541 (13.74)	-	-	.032 (0.81)		
47 2/							-	-	.279 (7.09)	.026 (0.66)		
48 2/							-	-	.309 (7.85)	.026 (0.66)		
49 2/							-	-	.341 (8.66)	.026 (0.66)		
50 2/							-	-	.479 (12.17)	.026 (0.66)		
51 2/							-	-	.509 (12.93)	.026 (0.66)		
52 2/							-	-	.541 (13.74)	.026 (0.66)		

1/ Dimensions are in inches. Metric equalvents are given for general information only.
2/ Contact termination type - .024 diagonal post.
3/ Contact termination type – wire hole.
4/ Contact termination type - .020 square post.

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REQUIREMENTS

Dimensions and configurations: See figure 1.

Material and finish:

Contact: 0.020 inch thick phosphor bronze in accordance with ASTM B139, composition “A”. Gold plate in accordance with ASTM B488, type II, code D, class 1.27, over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 30 to 100 microinches.

Insulator: ASTM D5948, type SDG-F.

Keying hardware: 2 each hex nuts and washers, stainless steel, class 300 in accordance with ASTM A582, passivated in accordance with SAE-AMS-QQ-P-35, and 2 each keying pins, free machining brass in accordance with ASTM B16, UNS C36000 or ASTM B121, UNS C35300 or C34200 or ASTM B124, UNS C37700 or ASTM B36 UNS C23000 or C24000 or C26000 or C26800, nickel plated in accordance with SAE-AMS-QQ-N-290. For direct government procurement the keying hardware shall be assembled to the connector. For other orders the keying hardware may be furnished loose as specified in contract or order.

Contact identification: The first and last contact in each row shall be identified by an alphabetical-numerical code (e.g., A01, B12). The letter designates the row and the number designates the contact position within that row. Every fifth contact in each row shall have its identification mark molded in.

Contact retention: 6 pounds.

Mating and unmating: The maximum insertion force, in pounds, shall not exceed a value equal to .5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of .11 times the number of contacts and shall not exceed the measured insertion force.

Contact separation force: 1 ounce minimum.

Contact resistance: The contact resistance shall not exceed 20 milliohms.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms, 60 Hz.

High altitude: 300 volts rms, 60Hz.

Current rating, maximum: 5 amperes.

Keying pin: When required, can be easily keyed to the desired positions by using a standard type wrench. Connectors are supplied with keying sockets in “1” and “A” positions.
Mating connectors: Shall conform to MIL-DTL-55302/90 and MIL-DTL-55302/92, or MIL-DTL-55302/94 and MIL-DTL-55302/96.

Part or Identifying Number (PIN): M55302/91-(dash number from figure I).

Group submission: MIL-DTL-55302/89, MIL-DTL-55302/92, and MIL-DTL-55302/95.

Previous changes. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/89
MIL-DTL-55302/90
MIL-DTL-55302/92
MIL-DTL-55302/94
MIL-DTL-55302/95
MIL-DTL-55302/96
ASTM A582
ASTM B16
ASTM B121
ASTM B124
ASTM B139
ASTM B36
ASTM B488
ASTM D5948
SAE-AMS-QQ-N-290
SAE-AMS-QQ-P-35

CONCLUDING MATERIAL

Custodians:
Army – CR
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-4440-010)

Review activities:
Army – AR, AT, AV, CR4, MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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